

Rachel Hurley

PhD Candidate WPI Worcester, MA | [LinkedIn](#) | rhurley@wpi.edu | [Website](#) | [Google Scholar](#)

RESEARCH INTERESTS

Indoor environmental quality, indoor wildfire smoke transport, material science, indoor occupant behavior, and indoor smoke damage remediation.

EDUCATION

Worcester Polytechnic Institute (WPI) | Worcester, MA
PhD, Civil & Environmental Engineering | Aug 2023 – Present
GPA: 4.0

Lafayette College | Easton, PA
B.S. in Integrative Engineering, Environment and Energy | Graduated May 2023
Honors: Co-Valedictorian, GPA: 4.0

RESEARCH EXPERIENCE

Research Assistant | Worcester Polytechnic Institute | Sep 2023 - Present

- Research focuses on indoor air quality and human decision-making impacted by wildfire smoke.

NSF GEOHEALTH Intern | Aerodyne Research | Sep 2024 – Mar 2025

- Learned chemical ionization mass spectrometry, study smoke re-emission from indoor materials.

Home Ignition Zones Researcher | San Jose State University | June 2024 – Aug 2024

- Studied homeowner preparedness for wildfire protection in California, responsible for recruitment, data collection, and data quality across three regions.

Undergraduate Research Assistant | Worcester Polytechnic Institute | June 2021 – May 2023

- Led two studies related to indoor health and cognition, led study design, data collection, analysis.

Claire Booth Luce Research/Excel Scholar | Lafayette College | Aug 2020 – May 2023

- Led analysis for undergraduate research project to predict flash flooding in the mid-Atlantic.

FELLOWSHIPS & AWARDS

National Science Foundation (NSF) Graduate Research Fellowship | Mar 2023

ASHRAE Grant-in-Aid Award (Life Member Club for Top 2 Applicants) | May 2025

Crommelin-Armiger Scholarship Fund Award | Nov 2025

Emerging Research Conference Student Travel Award | Jan 2025

American Association Aerosol Research Student Travel Award | Jan 2025

ISIAQ Indoor Air Conference Student Poster Award (Honorable Mention) | Jul 2024

WPI Duane D. Pearsall Endowment Innovation Jam Finalist | Apr 2024

UNDERGRADUATE HONORS

Landis Center Community Based Research Award | Lafayette College | Spring 2023

William C. McLean Tau Beta Pi Prize | Lafayette College | Spring 2021

Eugene P. Chase Phi Beta Kappa Prize | Lafayette College | Spring 2021

Marquis Scholar | Lafayette College | 2019 – 2023

PUBLICATIONS

- Hurley, R., Mangoubi, O., & Liu, S. (2026) Indoor wildfire smoke perceptions “nudge” mitigation behaviors. *Indoor Environments*, 3, 100150 [DOI].
- Hurley, R., Mangoubi, O., Zhang, K., & Liu, S. (2025). Impact of mask-wearing-related CO₂ exposure on performance – A global survey. *Building and Environment*, 285, 113537 [DOI].

- Hurley, R. F., Belyamani, M. A., Djamassbi, S., Somasse, G. B., Strauss, S., Zhang, H., Zhang, J. (Jensen), & Liu, S. (2024). High CO₂ exposure due to facemask wear is unlikely to impair cognition even in a warm environment after a long-term adaptation. *Energy and Built Environment*, S2666123324000655 [DOI].
- Hurley, R., Brandes, D., & Kelleher, C. (2024). Drivers and Buffers of Stream Flashiness in the Mid-Atlantic United States. *Hydrological Processes*, 38(10), e15302 [DOI].
- Belyamani, M. A., Hurley, R. F., Djamassbi, S., Somasse, G. B., Strauss, S., Zhang, H., Smith, M. J., Van Dessel, S., & Liu, S. (2024). Local wearable cooling may improve thermal comfort, emotion, and cognition. *Building and Environment*, 254, 111367 [DOI].

PUBLICATIONS IN PREPARATION

- Hurley, R., Avery, A., Simeoni, A., Canagaratna M., Liu, S., Bi, C. Interior Paint Alters the VOC Emission Behaviors in Wildfire-Smoke-Contaminated Drywalls (In preparation, EST Letters)
- Hurley, R., Saini, N., Schnider, C., Rosbach, D., Liu, S. The built environment and financial barriers drive older populations' wildfire smoke experience (In preparation, IJDRR)

CONFERENCE PRESENTATIONS

Podium Presentations

- Hurley, R., Liu, S., Avery, A., Simeoni, A., Canagaratna M., Bi, C. "Indoor Building Materials as Reservoirs for Prolonged Wildfire Sourced VOC Exposure" *AAAR*, Buffalo NY, 2025.
- Hurley, R., Bi, C., Liu, S. "Are Building Materials Reservoirs for Wildfire Sourced VOC Re-emissions and Indoor Exposure?" *AAAS ERN Conference in STEM*, Atlanta, GA, 2025.
- Hurley, R., Mangoubi, O., Liu, S. "Global Survey of the Impacts of Facemask Wear on Perceived Learning or Working Performance During COVID-19." *Indoor Air*, Honolulu, HI, 2024.
- Hurley, R., Brandes, D., Kelleher, C. "Comparison for Metrics of Stream Flashiness in the United States." *World Environmental and Water Resources Conference*, Henderson, NV, May 2023.
- Hurley, R., Amine, M. B., Liu, S. "Impact of Elevated CO₂ Exposure Due to a Face Mask on Thermal Comfort and Cognitive Performance in a Warm Enviro." *COBEE*, Montreal, Jul 2022.

Poster Presentations

- Hurley, R., Bi, C., Avery, A., Simeoni, A., Canagaratna M., Liu, S. "Re-Emission of Wildfire VOCs from Indoor Drywall Materials." *ASHRAE Winter Meeting*, Las Vegas, NV, 2026.
- Hurley, R., Mangoubi, O., Liu, S. "Health Symptoms and Mitigation Behaviors Are Associated with Olfactory Perception of Indoor Wildfire Smoke." *Indoor Air*, Honolulu, HI, 2024. (Honorable Mention)
- Hurley, R., Amine, M. B., Liu, S. "Impact of Mask-Wearing and Elevated CO₂ in the Breathing Zone on Cognitive Performance in a Warm Environment." *Indoor Air*, Honolulu, HI, 2024.

Seminar

- Hurley, R. "Leading the Charge Against Extreme Weather: Using Ashrae Standards to Enhance IAQ Resilience during Wildfire Smoke Events." *Seminar 59: Indoor Environmental Quality for Health During Extreme Weather Events. ASHRAE Winter Meeting*, Las Vegas, NV, 2026.

REPORTS

Final Report Editorial Committee Member | Lafayette College | Nov – Dec 2022

- Presented Integrated Plan for Achieving Carbon Neutrality to college President ([link](#)).

LEADERSHIP & SERVICE

ISIAQ Student Representative | International Society of Indoor Air Quality & Climate | Jul 2024 - 2026

- Elected position, strengthen student community engagement with programs, webinars, and attending board meetings. Reinstated mentorship program and organizing research competition.

Autumn School Coordinator | Healthy Building Asia Conference 2025 | Jul 2024 - Aug 2024

- Lead for schedule and activity planning for the 2-day student Autumn School event in India.

Summer School Student Coordinator | Indoor Air Conference 2026 | Sept 2026 - Jun 2026

- Lead for schedule and activity planning for student summer school preceding Indoor Air 2026.
- **Technical Committee Member** | ASHRAE | Mar 2024 – Present
- Member TC 7.10: Occupant Behavior in Building Design and Operation.
- Member TC 2.3: Gaseous Air Contaminants and Gas Contaminant Removal Equipment

Engineering Dean's Advisory Board | Worcester Polytechnic Institute | Spring 2026 – Present

- On panel of nominated students to work with Dean of Engineering to improve student experience.

President, Engineering Student Council | Lafayette College | Spring 2021 – Spring 2023

- Led division activities and planning, creating framework to strengthen community and belonging.

Panelist, AAUW Easton Girls Recognition Night | Easton, PA | May 2023

- Nominated by faculty to serve as a panelist to encourage middle school girls' interest in STEAM.

Journal Reviewer

- Building and Environment, Thinking Skills and Creativity.

MEDIA AND OTHER ACTIVITIES

WPI Podcast | Featured Guest | Oct 2025 "Wildfire Research Part Two" ([link](#))

NCAA Division III Athlete, Cross Country/Track & Field | WPI | Fall 2023 – Present

- Varsity Cross Country National Qualifying Team; 2023 Cross and 2024 Indoor 5k All-Region.

NCAA Division I Athlete, Cross Country/Track & Field | Lafayette College | Fall 2019 – Spring 2023

- 2019 MVP Freshman; All-Time Top 10 in 3000m steeple & 5000m; Patriot League Scorer.

Member, Oaks Leadership Academy | Lafayette College | Fall 2020 – Spring 2023

- Nominated by coaches for a selective leadership development program for student-athletes.